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Group 18 – Bus Delays in Greater Manchester

Project Info File

29 April 2026

Project Output Table

<i>Project Output</i>	<i>Output Description</i>
Project Output Files	Zip File Submitted to Moodle
Project Website	https://piyapa-uss.github.io/CASA0029-Group-18-Bus-Delay-in-Manchester/
Project Methodology Summary	In “Project Info & Methodology” Word Document

Individual Contributions Table

<i>Task Name</i>	<i>Major Contributors</i>	<i>Use of AI Tools in this Task</i>
Concept Development	All team members	No AI use
Website Design, User interface, Layout and Styling	Piyapa Sotthiwat	ChatGPT used for set-up website structure and code debugging
Overview/Research Framing	Piyapa Sotthiwat	ChatGPT used for code debugging
Accessibility & Bus Network Context	Kongtup Wanichjaroenporn	Claude was used to migrate the codebase from a standalone HTML/Mapbox prototype into the larger website as integrated JavaScript modules, converting Mapbox tilesets into compact local JSON files. It also helped debug, comment, and optimise the visualisation.

Delay & Reliability Dashboard	Kongtup Wanichjaroenporn	Claude is used for debugging, commenting, and optimising the visualisation.
Impact/Uneven Mobility	Jacob Echele	ChatGPT used to assist code debugging
About/Data	Jacob Echele	ChatGPT used to assist code debugging
Website Development	All Team Members	

Project Methodology Summary

Overview

Data. The overview draws on *Passenger journeys on local bus services by local authority, England*, published by the UK Department for Transport, restricted to five English combined-authority regions: Greater Manchester, West Midlands, West Yorkshire, Merseyside, and South Yorkshire. A boundary layer of English regions provides a small reference map alongside the chart.

Processing. Annual journey counts were aggregated from local-authority totals to these five city-regions and reshaped into a long CSV (year, area, journeys) consumed directly by D3. The regional boundary layer was simplified in QGIS to reduce geometric complexity while preserving overall spatial structure, yielding a lightweight GeoJSON suited to in-browser rendering.

Visualisation. The macro chart is a D3.js stacked area in which each region is a band and Greater Manchester is foregrounded through colour, opacity, and stroke weight. A year slider drives a clipped reveal of the stack and a vertical year marker, and a tooltip reports the selected year's journey count and Greater Manchester's rank. An inset SVG choropleth, rendered from the simplified GeoJSON via d3.geoMercator, acts as a geographic key.

Accessibility

Data. The accessibility section uses TfGM's GMAL (Greater Manchester Accessibility Levels) grid releases for February 2016 and February 2026. They are two snapshots of a 100 m grid covering 129,109 cells across Greater Manchester. Each cell contains scores for overall accessibility and for the four modes: Bus, Rail, Metrolink, and Local Link.

Processing. Raw shapefiles (in British National Grid, EPSG:27700) were read directly from their ZIP archives using GeoPandas. Each polygon cell was reprojected to WGS84 (EPSG:4326) and reduced to its representative point. Rather than exporting full GeoJSON, which would be

significantly larger, each cell was serialised as a compact fixed-position tuple [lon, lat, overall, bus, rail, metro, locallink], alongside a pre-computed summary statistics block (count, min, max, mean). This produces two compact JSON files, one per year, that can be loaded directly in the browser without a tile server.

Visualisation. The 3D map uses MapLibre GL for the basemap and deck.gl's ColumnLayer to render the grid as height-encoded columns. The column height represents the GMAL score, and the colour encodes either the 8-level GMAL palette (for overall accessibility) or a continuous diverging gradient (for individual modes). A year toggle switches between 2016 and 2026 data because both files contain cells in the same order, and per-cell changes are directly computable. A borough comparison heatmap, built from a point-in-polygon spatial join (GeoPandas sjoin) aggregated to Local Authority District level, provides a side-by-side ranking across all five metrics for both years, with a separate delta panel showing change.

Bus Delay

Data source. Delay analysis draws on static GTFS feeds and real-time GTFS-RT vehicle position records collected from 18th to 27th June 2025 within the study area.

Processing. The pipeline runs in four stages. Firstly, static feeds across all collected dates are unioned and filtered to the study area. Bus stop locations are snapped to OSM nodes, and A* routing is used to reconstruct per-route geometries, cached for reuse. Secondly, per-day GTFS-RT binary files are processed. Vehicle pings are snapped to route geometries and interpolated against scheduled stop times to produce per-arrival delay records, each tagged with a reliability flag (interpolated, dwell, or extrapolated). Thirdly, arrivals are aggregated across all dates by route, direction, stop, day type (weekday/weekend), and five time periods (AM off-peak, AM peak, midday, PM peak, PM off-peak), computing delay quantiles, on-time percentage, and late percentage. Finally, a dashboard export stage filters to bus routes with sufficient observations ($\geq 5,000$ arrivals per route), simplifies geometries using Douglas–Peucker, and computes route-level summary statistics as weighted medians of per-stop medians.

Visualisation. The delay dashboard uses Leaflet for a 2D rendering footprint. Routes are coloured by median delay across three threshold classes (≤ 2 min, 2–5 min, > 5 min). A heatmap overlay fills local areas by median delay, with opacity modulated by stop coverage so data-sparse areas fade rather than mislead. A time-of-day chip filter recolours all layers simultaneously. Clicking a route or LSOA cell drills into a stop-level detail view, showing per-stop delay distributions across all time periods.

Impact/Uneven Mobility

Data: Kongtup's bus delay dataset, which included data such as median delay, vehicle count, and bus stop geometries, was used in combination with a dataset from the UK government on the 2025 English Indices of Deprivation (IoD) Index, which assigns a composite score accounting for multiple deprivation metrics to each LSOA, marking that LSOA's level of deprivation.

Processing: In R, the bus delay dataset from the earlier section was read in and ensured to have the correct CRS, EPSG 4326. Stops with a recorded vehicle count of 0 were excluded to filter out stops with no active bus arrivals. Following this, a mean delay time was calculated per LSOA and weighted by vehicle count so that busier LSOAs are proportionally accounted for. Then, the data was joined with each LSOA's deprivation score from the 2025 English IoD dataset. Using these values, a deprivation percentile measure was calculated for each LSOA using the `dplyr` package's `percent\_rank` function, subtracted from one, then multiplied by 100 to create a percentage to help contextualise the deprivation scores for unfamiliar audiences. Higher percentiles represent areas with lower deprivation scores. The new, combined dataset was then exported to a GeoJSON format.

Visualisation: The dataset was then mapped using Mapbox, detailing one map for average delay, one for deprivation score, and one for the deprivation percentile of each LSOA, and graphed with the help of [chart.js](#). Two histograms were implemented, one to show how often each stop in the chosen LSOA experienced delays of different lengths and one to show how the average delay in the LSOA compares with Greater Manchester as a whole. Bin sizes are set to indicate the length of each delay.